Resource Allocation Playbook for Flood Relief Forecasting

Predictive Analytics for Resource Allocation in Disaster and NGO Operations, Coastal Odisha Districts

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Live dashboard, run locally. After starting the app, open http://localhost:8502/ in your browser. In a terminal, move into the dashboard folder (the path contains spaces, so wrap it in quotes), then run the command "streamlit run app.py --server.port 8502". Keep that terminal window open while using the dashboard.

Purpose

This playbook describes how an NGO operations team can translate the weekly water demand forecasts produced by the Prophet, LSTM, and ensemble models into concrete pre positioning and resupply decisions for six coastal Odisha districts, namely Puri, Ganjam, Kendrapara, Balasore, Bhadrak, and Jagatsinghpur.

Weekly Operating Cadence

- Every Sunday evening, refresh the input data with the latest IMD rainfall bulletins, river gauge readings, and cyclone advisories for each district.
- Run the one step ahead ensemble forecast to generate the predicted water demand in litres for the coming week, for all six districts.
- Convert the water demand forecast into a full relief package using the saved per litre resource ratios for dry ration kits, ORS units, tarpaulins, and medical kits.
- Review the alert label assigned to each district, which is HIGH, MODERATE, or LOW, based on the forecast crossing the 1,000 litre actionable threshold and the 10,000 litre high alert threshold.

Decision Rules by Alert Level

- HIGH alert, forecast at or above 10,000 litres. Treat as a near certain resupply event. Begin pre positioning of water, ORS, dry rations, tarpaulins, and medical kits within 48 hours, ahead of the forecast week, using the resource ratio conversion. Notify district disaster authorities and confirm warehouse to last mile transport readiness.
- MODERATE alert, forecast between 1,000 and 10,000 litres. Place the district on a watch list. Pre stage a partial buffer stock, roughly 30 to 50 percent of the HIGH alert quantity, and recheck the forecast at the next weekly cycle before committing full resources.
- LOW alert, forecast below 1,000 litres. No special action beyond routine stock replenishment. Continue normal monitoring so that a district moving from LOW to MODERATE or HIGH in successive weeks is caught early.

Translating Forecasts into Supply Quantities

For each district, multiply the forecast water demand in litres by the saved resource ratios to obtain the recommended quantities of dry ration kits, ORS units, tarpaulin units, and medical kits. These ratios were derived as the median, across historical non zero demand weeks, of each resource divided by water demand, following SPHERE handbook style per capita bundling. This keeps the bundle proportions consistent with past relief operations while scaling automatically to the size of the predicted event.

Handling Uncertainty and Communicating with Coordinators

Because flood relief demand is highly intermittent, with a large share of weeks at zero and the remaining weeks spanning a very wide range, the dashboard avoids relying on a single noisy point forecast for the yes or no resupply decision. Instead, the alert accuracy metric, which checks whether the forecast correctly predicts crossing the actionable threshold, is the primary number a relief coordinator should trust. The point forecast in litres should be treated as a sizing guide for the pre positioned bundle, while the alert label, HIGH, MODERATE, or LOW, is the trigger for action.

Handling Data Gaps in Remote Districts

When recent rainfall, river gauge, or cyclone data is missing for a district, the pipeline falls back to the trailing four week average of each weather variable as a persistence assumption, and the LSTM continues to use the most recent twelve weeks of available features. Operations staff should flag any week where this fallback was used, so that coordinators apply additional judgement and, where possible, supplement the forecast with field reports from local volunteers before finalising pre positioning quantities.

Roles, Responsibilities, and Expected Impact

The data and forecasting lead refreshes inputs and reruns the pipeline weekly. The operations manager reviews alert levels each Monday and authorises pre positioning for HIGH and MODERATE districts. The warehouse and logistics team executes pre positioning within the 48 hour window, and field coordinators feed ground truth incident and supply data back into the dataset to improve future forecasts. Pre positioning supplies roughly 48 hours before a flood event can reduce relief delivery cost by around 30 percent, allowing the same budget to serve substantially more people, shifting the NGO from reactive logistics to proactive, forecast driven resource allocation across the six coastal Odisha districts.